



B.Tech. CSE VI CSS 338
EMBEDDED SYSTEMS LAB

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Roll Number: BTECH/10007/23

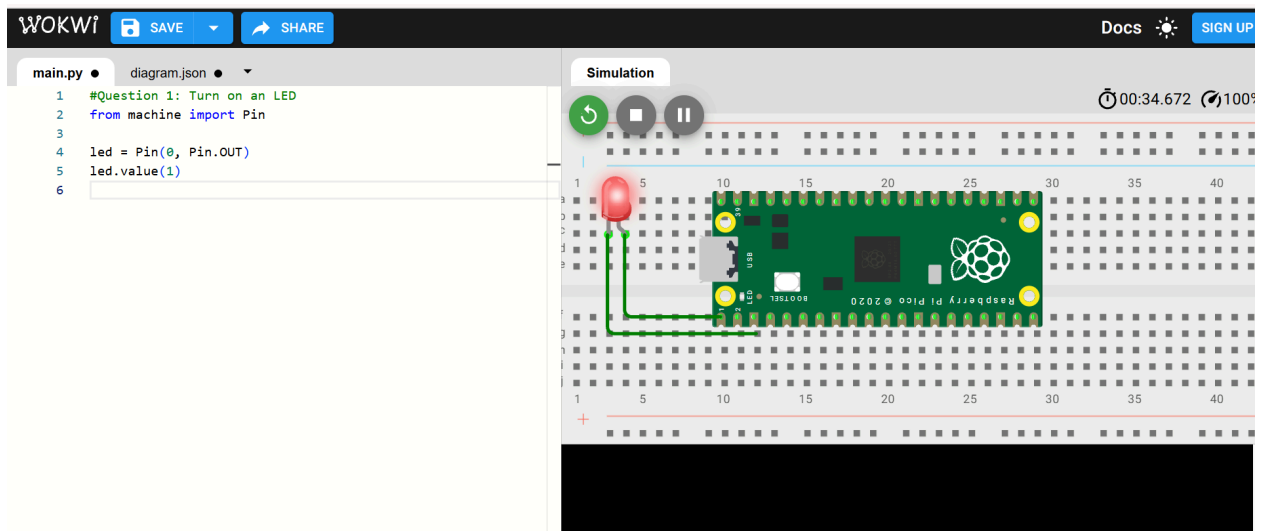
Assignment-1

Using Micro Python, simulate the following on a Raspberry Pi Pico:

1. Turn on an LED

Code

```
from machine import Pin
led = Pin(0, Pin.OUT)
led.value(1)
```

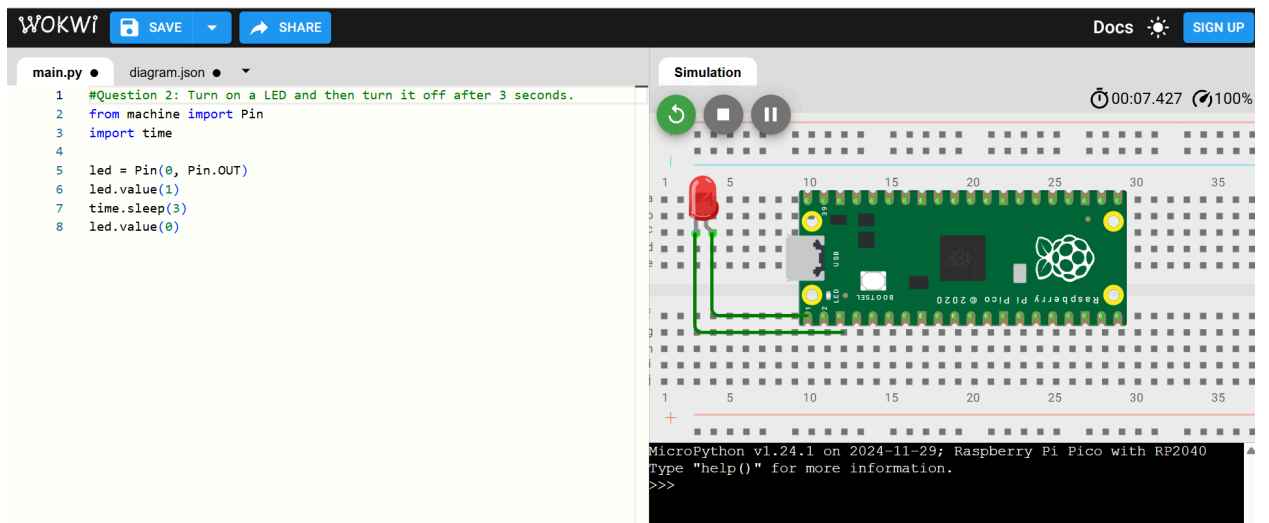
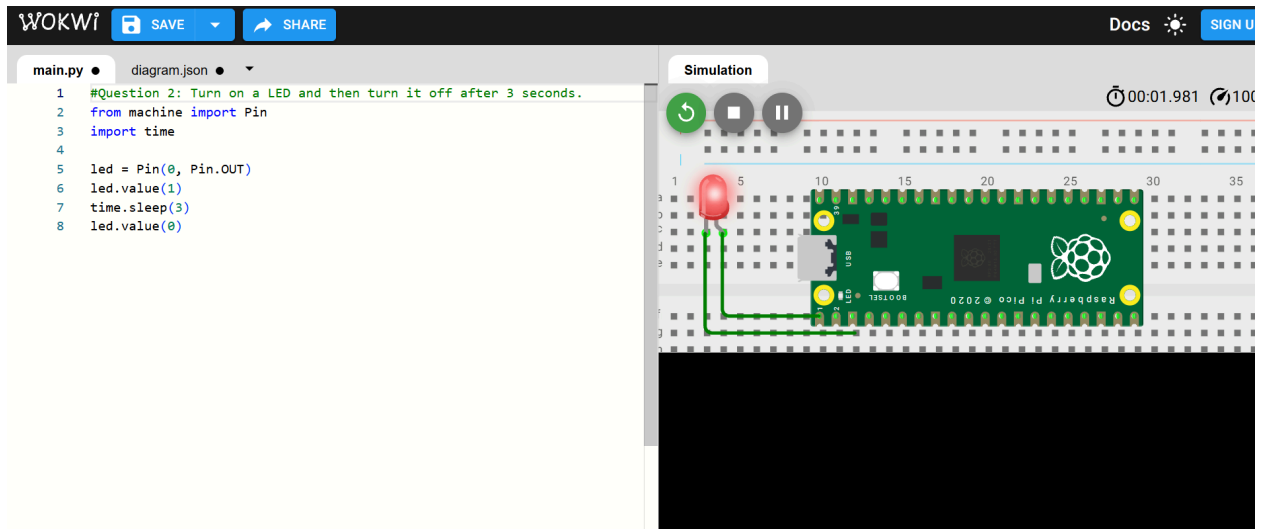


2. Turn on a LED and then turn it off after 3 seconds.

Code

```
from machine import Pin
import time

led = Pin(0, Pin.OUT)
led.value(1)
time.sleep(3)
led.value(0)
```



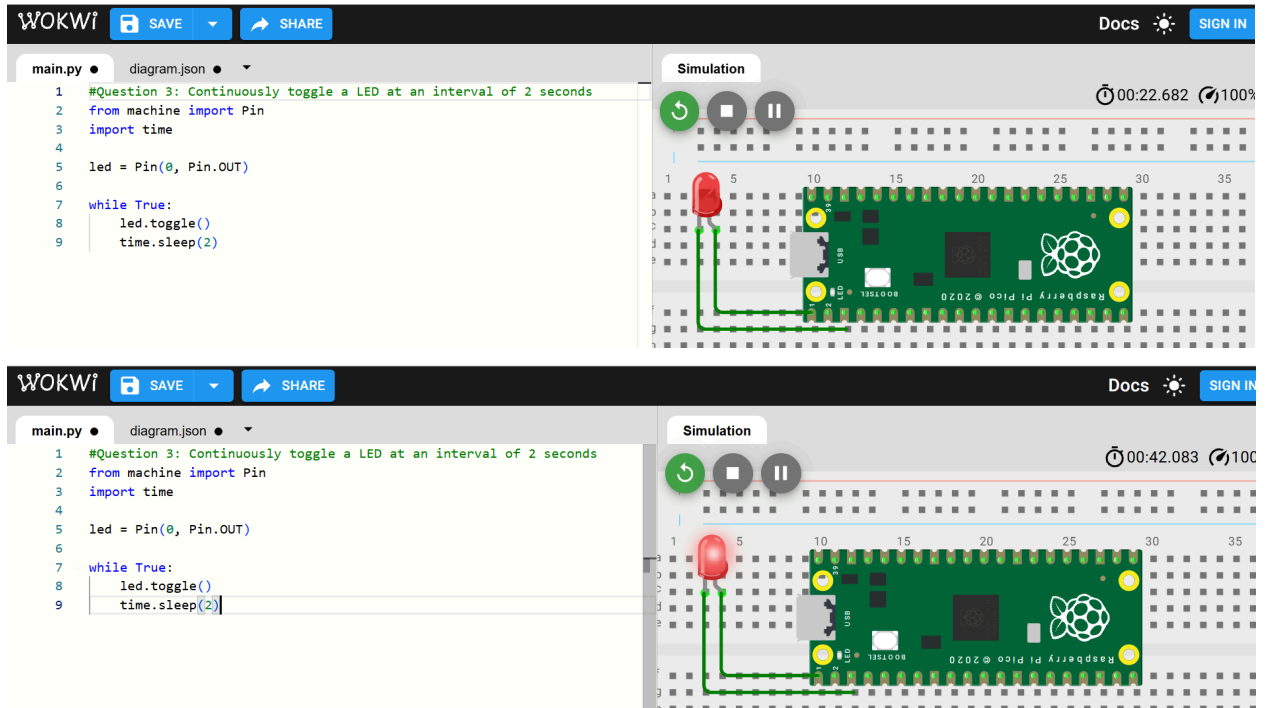
3. Continuously toggle a LED at an interval of 2 seconds

Code

```
from machine import Pin
import time
```

```
led = Pin(0, Pin.OUT)
```

```
while True:
    led.toggle()
    time.sleep(2)
```



4. Continuously toggle two pair of LEDs with one pair turning on and the other turnig off after intervals of two seconds

Code

#Question 4:

#Continuously toggle two pair of LEDs with one pair turning on

#and the other turnig off after intervals of two seconds

```
from machine import Pin
```

```
import time
```

```
pair1 = [Pin(0, Pin.OUT), Pin(1, Pin.OUT)]
```

```
pair2 = [Pin(2, Pin.OUT), Pin(3, Pin.OUT)]
```

```
while True:
```

```
    # Turn ON pair1, OFF pair2
```

```
    for led in pair1:
```

```
        led.value(1)
```

```
    for led in pair2:
```

```
        led.value(0)
```

```
time.sleep(2)
```

```
# Turn OFF pair1, ON pair2
```

```
for led in pair1:
```

```
    led.value(0)
```

```
for led in pair2:
```

```
    led.value(1)
```

```
time.sleep(2)
```

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main.py • diagram.json •

```
1 #Question 4:
2 #Continuously toggle two pair of LEDs with one pair turning on
3 #and the other turning off after intervals of two seconds
4 from machine import Pin
5 import time
6
7 pair1 = [Pin(0, Pin.OUT), Pin(1, Pin.OUT)]
8 pair2 = [Pin(2, Pin.OUT), Pin(3, Pin.OUT)]
9
10 while True:
11     # Turn ON pair1, OFF pair2
12     for led in pair1:
13         led.value(1)
14     for led in pair2:
15         led.value(0)
16     time.sleep(2)
17
18     # Turn OFF pair1, ON pair2
19     for led in pair1:
20         led.value(0)
21     for led in pair2:
22         led.value(1)
23     time.sleep(2)
24
```

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```
1 #Question 4:
2 #Continuously toggle two pair of LEDs with one pair turning on
3 #and the other turning off after intervals of two seconds
4 from machine import Pin
5 import time
6
7 pair1 = [Pin(0, Pin.OUT), Pin(1, Pin.OUT)]
8 pair2 = [Pin(2, Pin.OUT), Pin(3, Pin.OUT)]
9
10 while True:
11     # Turn ON pair1, OFF pair2
12     for led in pair1:
13         led.value(1)
14     for led in pair2:
15         led.value(0)
16     time.sleep(2)
17
18     # Turn OFF pair1, ON pair2
19     for led in pair1:
20         led.value(0)
21     for led in pair2:
22         led.value(1)
23     time.sleep(2)
24
```

Simulation 00:21.966 99%

5. Simulate a traffic light system the lights turn on Red, Orange and Green in sequence after waiting a random amount of time (measure in seconds) where the time is never more than 5 seconds and never less than two seconds.

Code

```
#Question 4:
from machine import Pin
import time
import random

led_pins = [0, 1, 2]
leds = [Pin(pin, Pin.OUT) for pin in led_pins]

while True:
    for led in leds:
        # Turn OFF all LEDs first
        for l in leds:
            l.value(0)

        # Turn ON current LED
        led.value(1)

        # Random delay between 2 and 5 seconds
        time.sleep(random.randint(2, 5))
```

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main.py

diagram.json

```

1  #Question 5:
2  #Simulate a traffic light system the lights turn on Red, Orange and Green
3  #in sequence after waiting a random amount of time (measure in seconds)
4  #where the time is never more than 5 seconds and never less than two seconds.
5  from machine import Pin
6  import time
7  import random
8
9  led_pins = [0, 1, 2]
10 leds = [Pin(pin, Pin.OUT) for pin in led_pins]
11
12 while True:
13     for led in leds:
14         # Turn OFF all LEDs first
15         for l in leds:
16             l.value(0)
17
18         # Turn ON current LED
19         led.value(1)
20
21         # Random delay between 2 and 5 seconds
22         time.sleep(random.randint(2, 5))
23

```

Simulation

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main.py

diagram.json

```

1  #Question 5:
2  #Simulate a traffic light system the lights turn on Red, Orange and Green
3  #in sequence after waiting a random amount of time (measure in seconds)
4  #where the time is never more than 5 seconds and never less than two seconds.
5  from machine import Pin
6  import time
7  import random
8
9  led_pins = [0, 1, 2]
10 leds = [Pin(pin, Pin.OUT) for pin in led_pins]
11
12 while True:
13     for led in leds:
14         # Turn OFF all LEDs first
15         for l in leds:
16             l.value(0)
17
18         # Turn ON current LED
19         led.value(1)
20
21         # Random delay between 2 and 5 seconds
22         time.sleep(random.randint(2, 5))
23

```

Simulation

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main.py

diagram.json

```

1  #Question 5:
2  #Simulate a traffic light system the lights turn on Red, Orange and Green
3  #in sequence after waiting a random amount of time (measure in seconds)
4  #where the time is never more than 5 seconds and never less than two seconds.
5  from machine import Pin
6  import time
7  import random
8
9  led_pins = [0, 1, 2]
10 leds = [Pin(pin, Pin.OUT) for pin in led_pins]
11
12 while True:
13     for led in leds:
14         # Turn OFF all LEDs first
15         for l in leds:
16             l.value(0)
17
18         # Turn ON current LED
19         led.value(1)
20
21         # Random delay between 2 and 5 seconds
22         time.sleep(random.randint(2, 5))
23

```

Simulation

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6. Simulate a two input truth table using three LEDs.

Code

```
#Question 6:
#Simulate a two input truth table using three LEDs.
from machine import Pin
import time

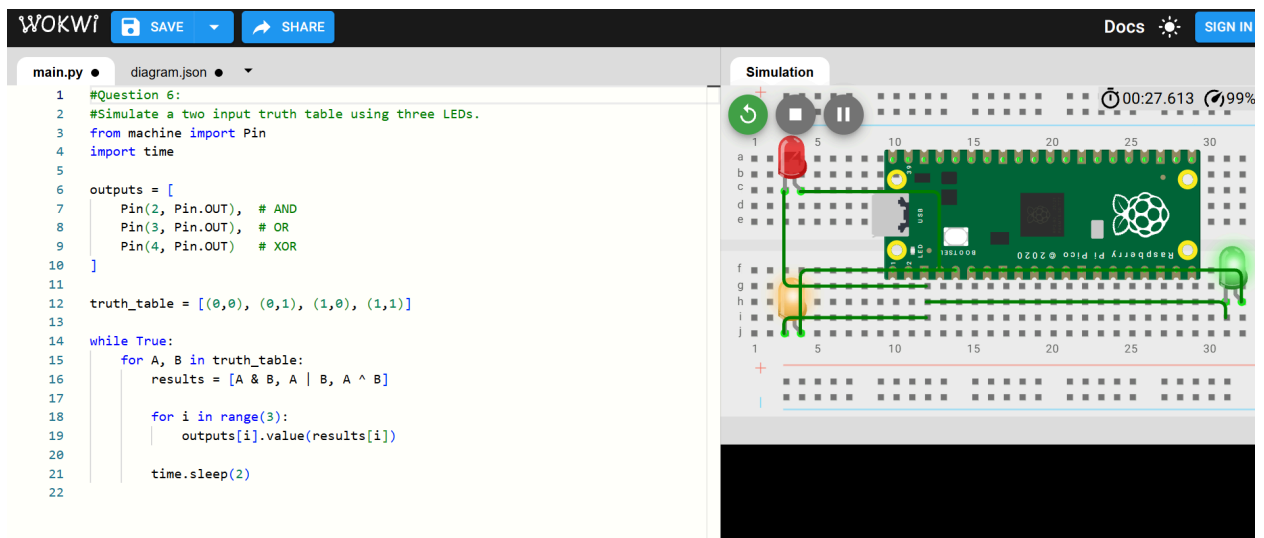
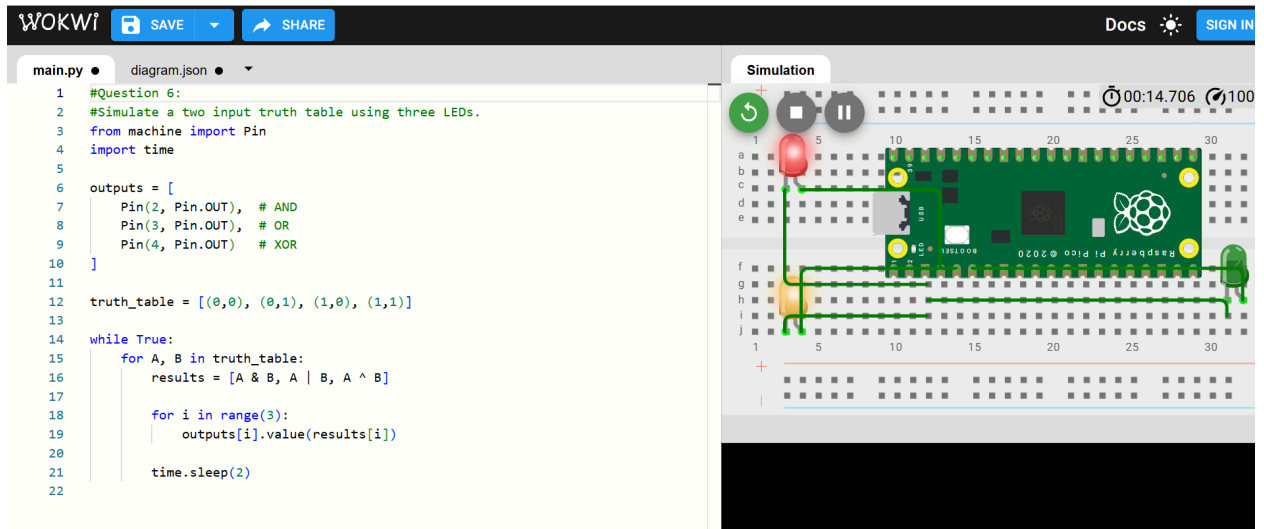
outputs = [
    Pin(2, Pin.OUT), # AND
    Pin(3, Pin.OUT), # OR
    Pin(4, Pin.OUT)  # XOR
]

truth_table = [(0,0), (0,1), (1,0), (1,1)]

while True:
    for A, B in truth_table:
        results = [A & B, A | B, A ^ B]

        for i in range(3):
            outputs[i].value(results[i])

        time.sleep(2)
```

7. Create a system of eight LEDs arranged in a circular fashion. Create a simulation by randomly turning on the LEDs in sequence. After completion of a complete round decrease the time between the LEDs turning on by an amount δ .

Code

#Question 7:

```

from machine import Pin
import time
import random

```

```

leds = [Pin(i, Pin.OUT) for i in range(8)]

```

```
while True:
    random.shuffle(leds)

    for led in leds:
        led.value(1)
        time.sleep(0.3)
        led.value(0)
```

8. **Create a square formation using LEDs of size 4 x 4. Turn on the LEDs individually passing row by row from top to bottom and then in the reverse direction continuously.**